

- 7 The points A, B, C have position vectors

$$2\mathbf{i} + 2\mathbf{j}, \quad -\mathbf{j} + \mathbf{k} \quad \text{and} \quad 2\mathbf{i} + \mathbf{j} - 7\mathbf{k}$$

respectively, relative to the origin O .

- (a) Find an equation of the plane OAB , giving your answer in the form $\mathbf{r} \cdot \mathbf{n} = p$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

The plane Π has equation $x - 3y - 2z = 1$.

- (b) Find the perpendicular distance of Π from the origin. [1]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (c) Find the acute angle between the planes OAB and Π . [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (d) Find an equation for the common perpendicular to the lines OC and AB . [10]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....